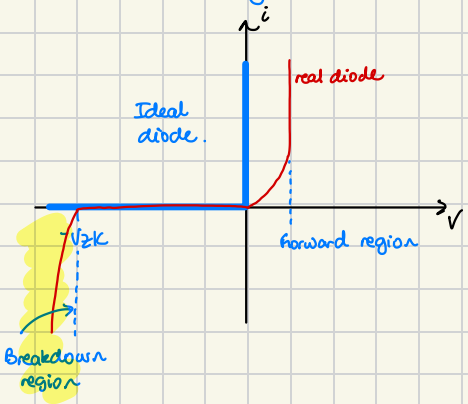
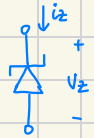
ELG 2136

Lecture 4

Zener diode

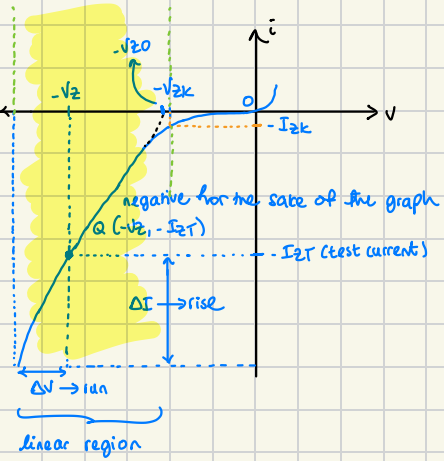
Zener diodes: diodes operating in the breakdown region.





$i_Z > 0$ [positive values but represent
$V_Z > 0$ [negative values on the i-v plot]

Magnifying the breakdown region



positive values by convention

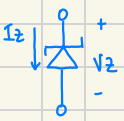
$Q(V_Z, I_{ZT})$

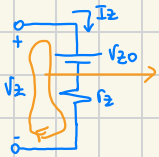
m = slope = $\frac{\text{rise}}{\text{run}} = \frac{\Delta I}{\Delta V}$. Ohm's law: $V = IR \Rightarrow \boxed{\frac{I}{V} = \frac{1}{R}}$

$\Rightarrow \boxed{m = \frac{\Delta I}{\Delta V} = \frac{1}{r_Z}}$ $\quad r_Z = \frac{1}{m}$

r_Z: zener diode resistance

Zener diode model in breakdown region





KVL: $-V_Z + V_{Z0} + I_Z r_Z = 0$

$\Rightarrow \boxed{V_Z = V_{Z0} + I_Z r_Z}$